

2016.0 RANGE ROVER (LG), 308-07

TRANSFER CASE - VEHICLES WITH: SINGLE SPEED TRANSFER CASE

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description of the Transfer Case - Single speed, refer to the relevant Description and Operation section in the workshop manual.

REFER to: [Transfer Case](#) (308-07B Transfer Case - Vehicles With: Single Speed Transfer Case, Description and Operation).

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.

1. Verify the customer concern

2. Visually inspect for obvious signs of damage and system integrity

Visual Inspection

MECHANICAL

- Transfer case
- Driveshafts
- Differentials
- Halfshafts
- Fluid leaks

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step

4. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index

5. Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Warning message/gearbox fault displayed on instrument cluster	Transfer case system fault	- Using the manufacturer approved diagnostic system, perform routine - Inline diagnostic unit 2 non-intrusive test -Transfer case - Pass - Recheck operation - Fail - Refer to the relevant DTC index
Whining noise	Transfer case bearing/chain wear	- Using the manufacturer approved diagnostic system, perform routine - Noise, vibration and harshness sensor diagnostic test - Transfer case - Refer to the relevant section of the workshop manual and install a new transfer case chain - Refer to the relevant section of the workshop manual and install a new transfer case rear output shaft bearing

JUDDER DURING LOW SPEED MANOEUVRING WITH HIGH STEERING INPUTS

At full lock, there will be some tyre scrubbing; as the temperature drops, scrubbing may be experienced as stick/slip judder. This is normal and does not require any service intervention

VIBRATION FROM TRANSFER CASE

Vibration from the transfer case may be caused by the installation of

incorrect front/rear drive units. Refer to Sections 205-02 and 205-03 for further guidance on how to diagnose and rectify this issue